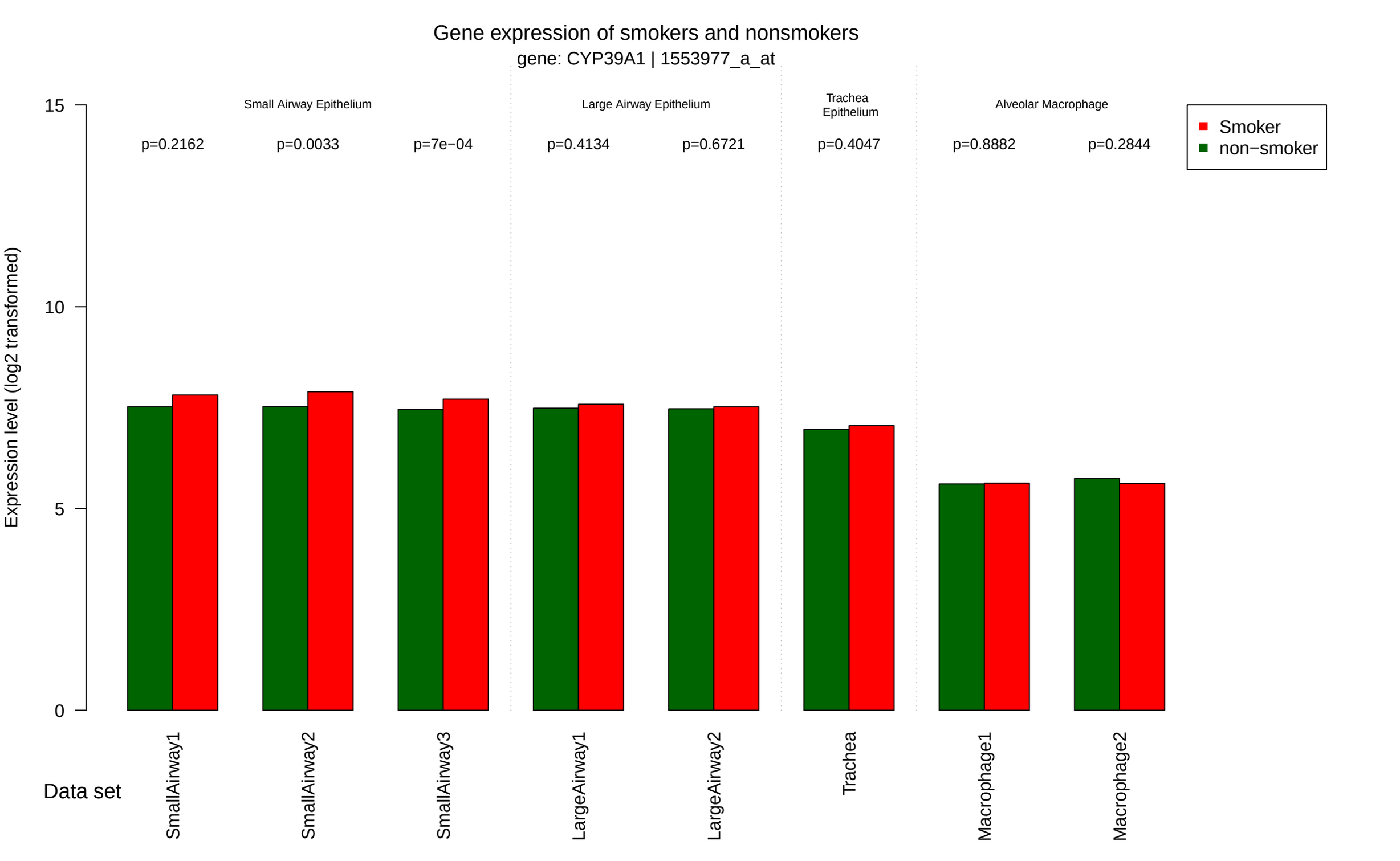
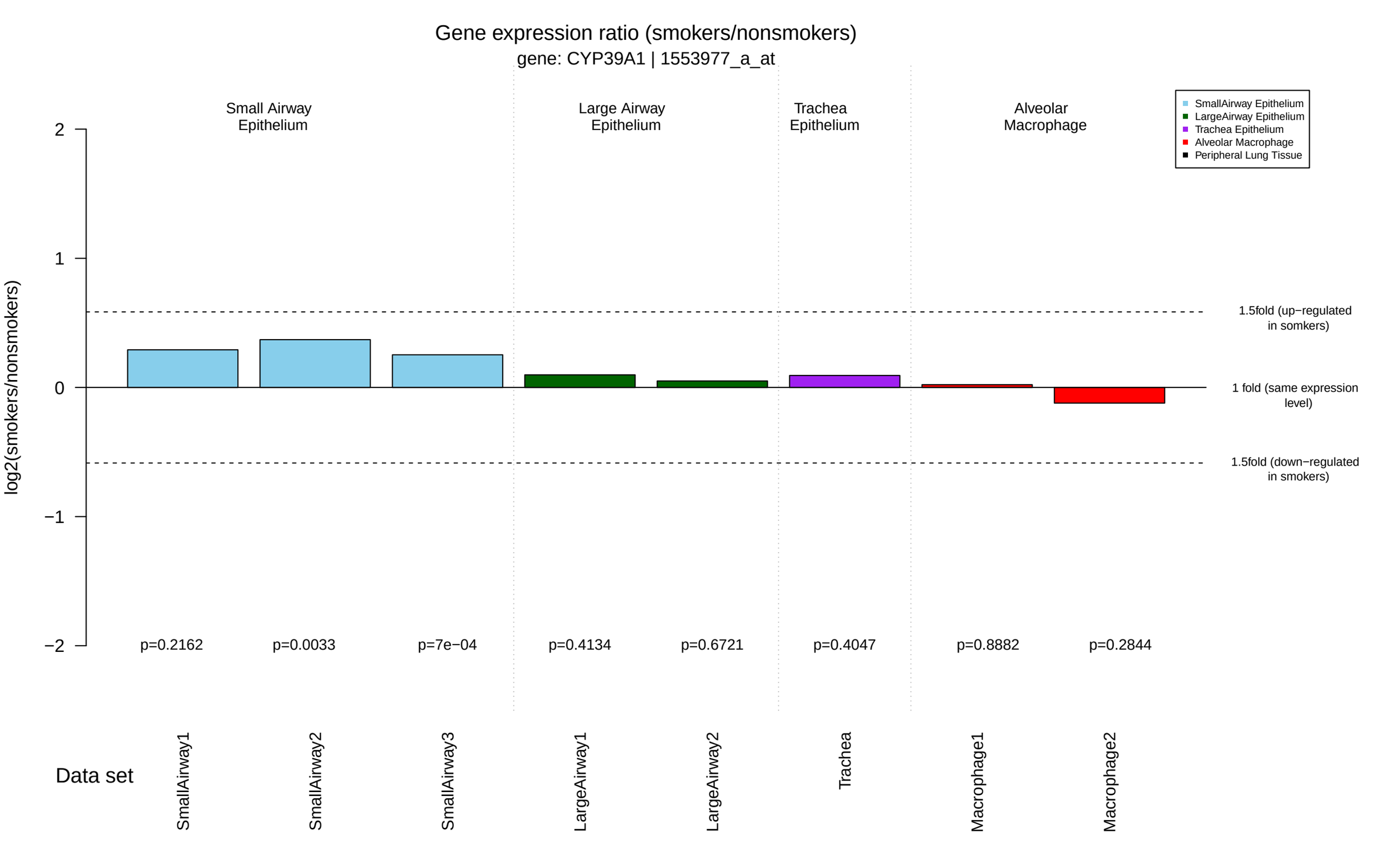
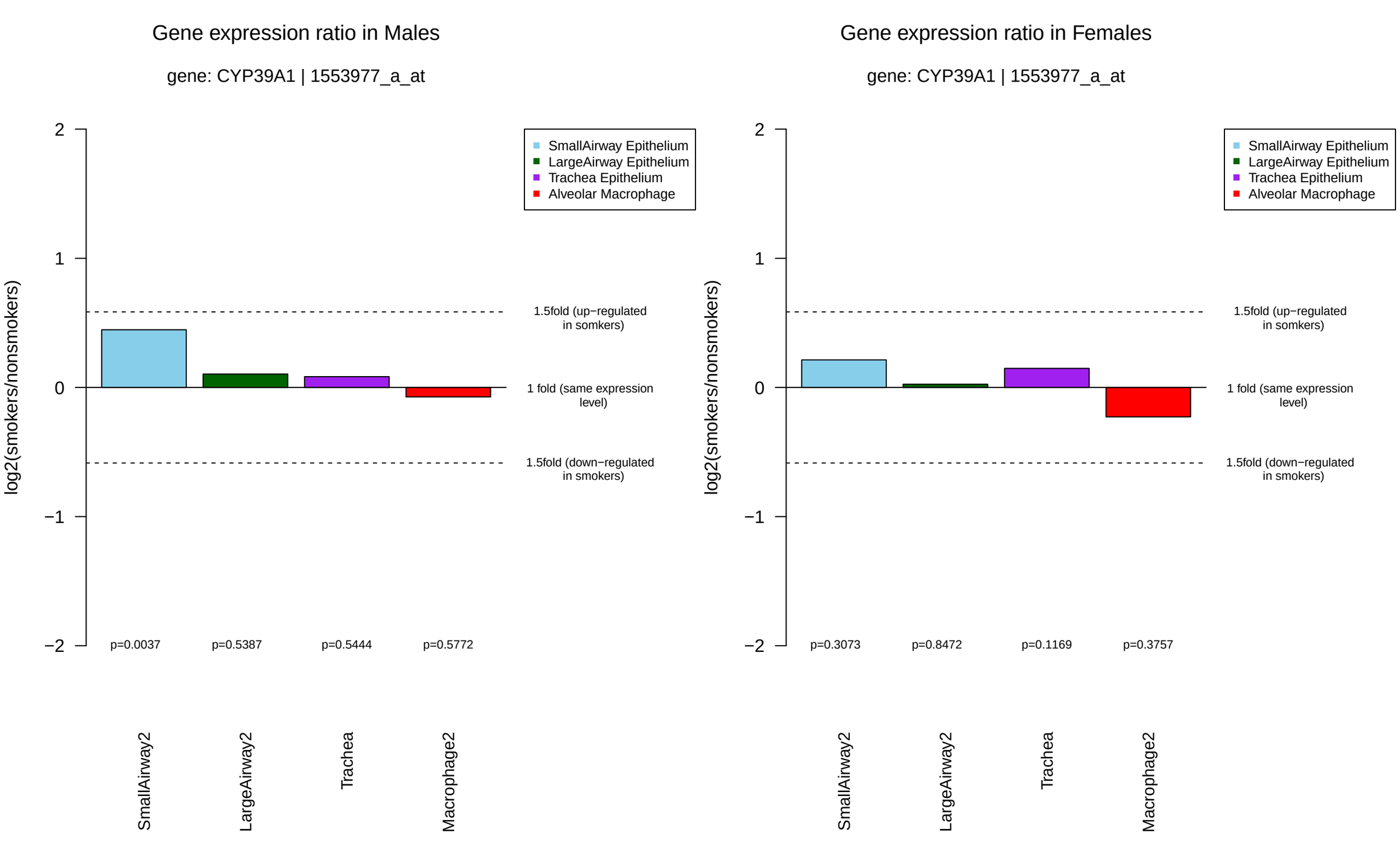
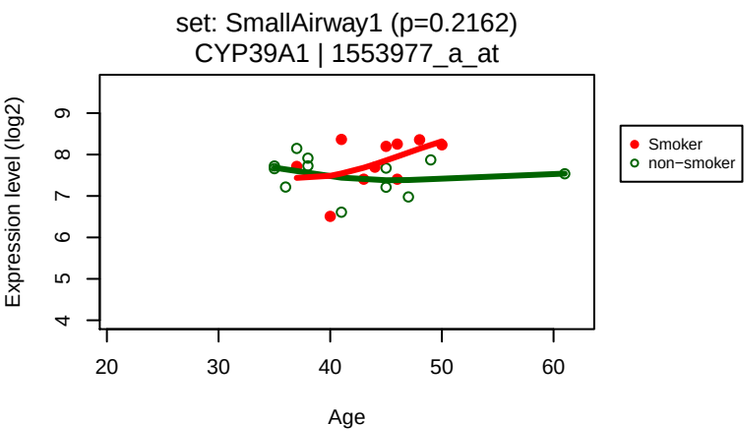
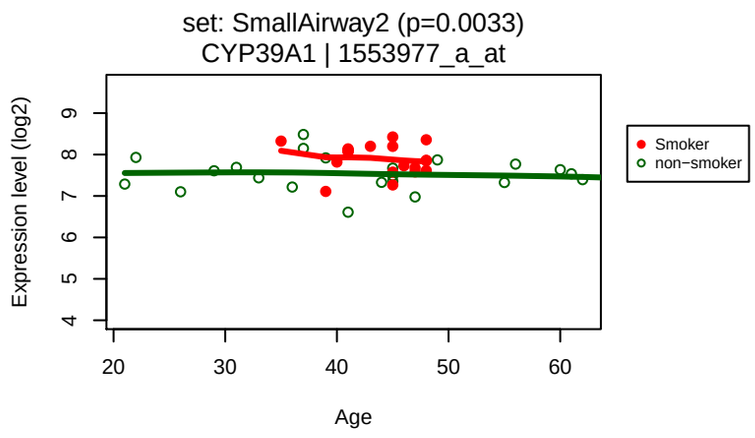
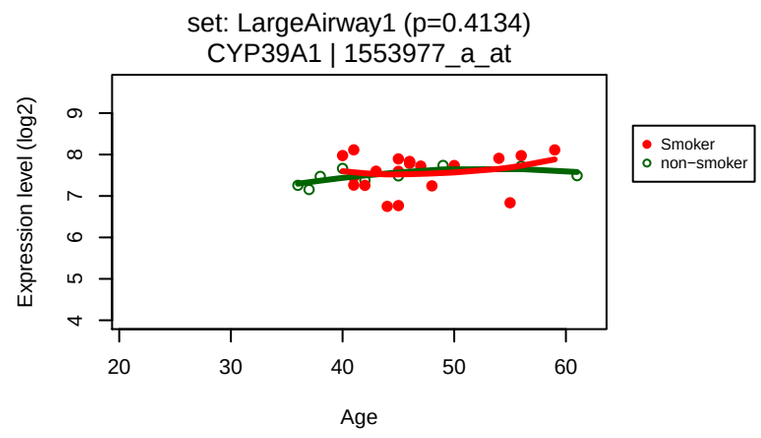

Gene expression of smokers and nonsmokers
gene: CYP39A1 | 1553977_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2162
p=0.0033
p=7e−04
p=0.4134
p=0.6721
p=0.4047
p=0.8882
p=0.2844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

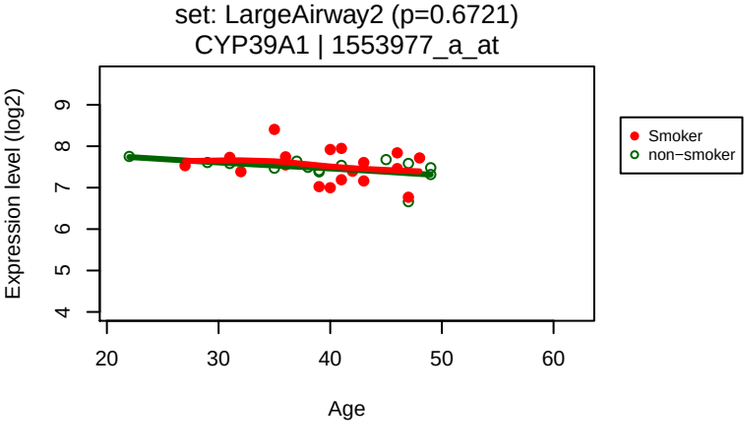

Gene expression ratio (smokers/nonsmokers)
gene: CYP39A1 | 1553977_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2162
p=0.0033
p=7e−04
p=0.4134
p=0.6721
p=0.4047
p=0.8882
p=0.2844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

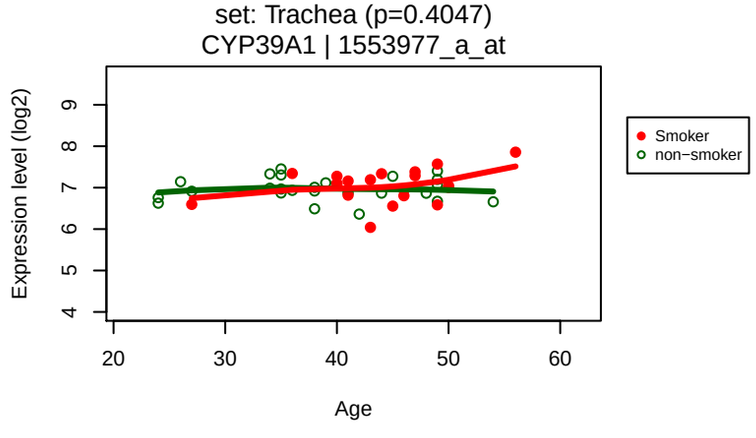

Gene expression ratio in Males
gene: CYP39A1 | 1553977_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.5387
p=0.5444
p=0.5772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP39A1 | 1553977_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3073
p=0.8472
p=0.1169
p=0.3757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

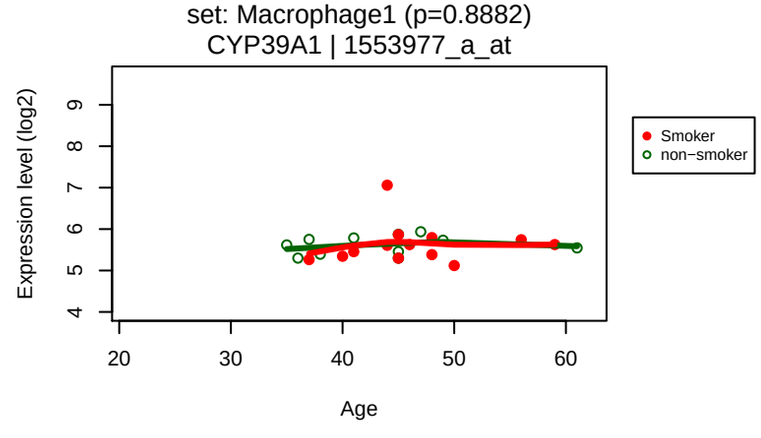

set: SmallAirway1 (p=0.2162)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker

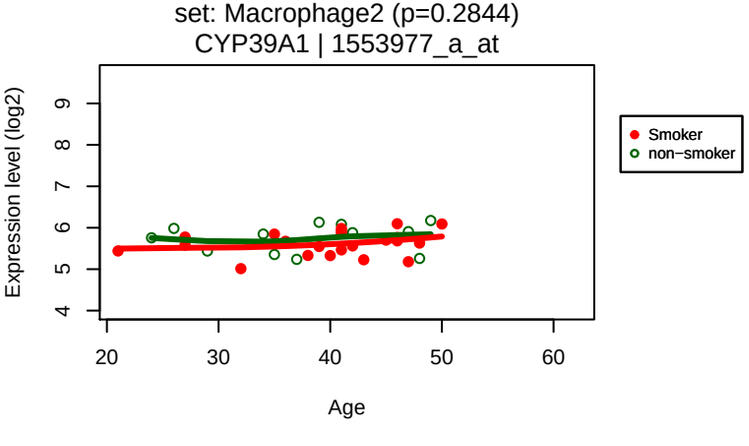

set: SmallAirway2 (p=0.0033)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker

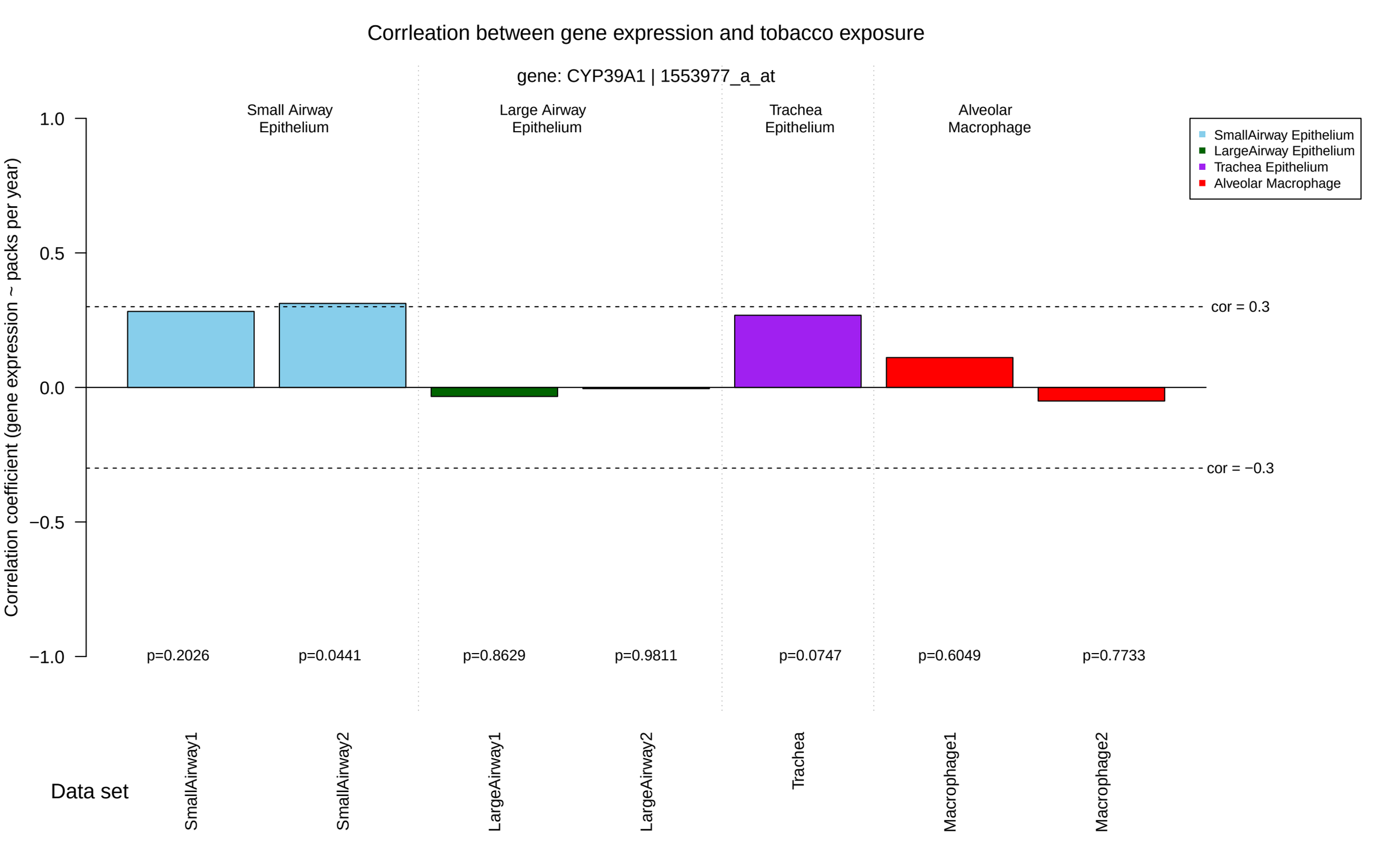

set: LargeAirway1 (p=0.4134)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6721)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4047)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8882)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2844)
CYP39A1 | 1553977_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP39A1 | 1553977_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2026
p=0.0441
p=0.8629
p=0.9811
p=0.0747
p=0.6049
p=0.7733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2